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Week 8 Quiz

Graded Quiz • 30 min

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Week 8 Quiz

Quiz: Week 8 QuizSubmitted

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The multi-layer Perceptron has to have at least two hidden layers. Choose the correct answer

DueAug 11, 11:59 PM IST

Attempts3 every 2 hours

1 / 1 point

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That is correct. It can be used for many, hidden layers.

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1. The multi-layer Perceptron has to have at least two hidden layers. Choose the correct answer

☐ True

☒ False

👍 Correct

That is correct. It can be used for many, hidden layers.

2. The main difference between a hidden layer unit (TLU) and a processing element (PE) is (choose one):

☐ Their outputs reach different values in the limit

☐ The TLU has fewer inputs than the PE

☒ The PE includes a differentiable activation function whereas the TLU does not

👍 Correct

That is correct. The differentiable activation function makes it possible to develop a training process for the neural network

3. The depth of e neural network us defined by (choose one):

☒ The number of hidden layers

☐ The total number of layers

☐ The number of hidden layers plus 1

👍 Correct

That is correct, and is the definition of network depth

4. A fully connected neural network (choose one):

☐ Ensures that there is at least one pathway from the input nodes to the output nodes

☐ Must have at least one hidden layer

☒ All outputs from the node in one layer are connected to all nodes of the subsequent layer

👍 Correct

That is correct because all possible connections are used.

5. A neural network has been trained to classify an image into 16 classes. The number of nodes in the output layer can be

☐ 5

☐ 7

☒ 4

👍 Correct

That is correct: 16 classes can be represented by 4 nodes if those nodes are a binary code (from 0 to 15, i.e. 16 cases)

6. The use of a momentum term during neural network training is designed to (choose one):

☒ improve convergence and speed up training

☐ Allow some of the less important linkages between layers to be removed

☐ Set a limit on the number of iterations to be used.

👍 Correct

That is correct.

7. The neural network is (choose one):

☐ a binary classifier

☒ A multi-class classifier

☐ A multi-class decision tree

👍 Correct

That is correct.

8. Design of a neural network for remote sensing applications requires the user to

☐ Specify the number of nodes to use in a single hidden layer

☐ Specify the number of hidden layers to use and the number of nodes in all layers

☒ Specify the number of hidden layers to use, the number of nodes in all layers and the nature of the activation function

👍 Correct

That is correct. These parameters fully specify the design

9. In its fundamental operation (based on its underlying equations) the neural network is most like (choose one):

☐ The maximum likelihood classifier

☐ The support vector machine

☒ The minimum distance classifier

👍 Correct

That is correct. It is based on many linear decisions.

10. Which of the following statements is true?

☐ Iteration over a labelled training set of pixels is required for both the neural network and the support vector classifier

☐ Both the maximum likelihood classifier and the neural network make use of class covariance matrix information

☒ The maximum likelihood classifier and the neural network are multi-class techniques.

👍 Correct

That is correct

https://www.coursera.org/learn/remote-sensing/exam/7gfoX/week-8-quiz/attempt/redirectToCover=true

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